

Phonons in EDUS: the equations solved for each combination of the input parameters

EDUS development notes, branch `ehrenfest`

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Three input parameters decide what the Ehrenfest dynamics of the lattice actually solves:

- the **mean field** of the electrons (`"coulomb": true`: Hartree with RPA, Hartree + SEX with HSEX) or IPA (`"coulomb": false`);
- `phonons/coupling`: `"screened"` (default, standard EPW run) or `"bare"`;
- `phonons/adiabatic_reference`: `"none"`, `"static"` (default) or `"dynamic"`.

The equations of motion have always the same form (section 1); the parameters only choose four ingredients: the coupling g in the Hamiltonian and in the force, the force constants K , the reference density matrix ρ_{ref} of the force, and how the electronic force constants Π are computed at the beginning. Section 3 lists them for the nine combinations; the following sections explain what each choice means. The code is in `src/Phonons/Lattice.cpp` (`initialize`, `unscreen_coupling`, `static_response`) and `src/Propagator/Propagator.cpp`.

1 Equations common to all the cases

$$i \dot{\rho}(k) = \left[H_0(k) + \mathbf{E}(t) \cdot \mathbf{r} + \Sigma[\rho - \rho_0] + \sum_{\mu} u_{\mu} g_{\mu}(k), \rho(k) \right], \quad (1)$$

$$M_{\mu} \ddot{u}_{\mu} = - \sum_{\nu} K_{\mu\nu} u_{\nu} - F_{\mu} - \frac{2M_{\mu}}{\tau} \dot{u}_{\mu}, \quad (2)$$

$$F_{\mu} = \langle g_{\mu}, \rho - \rho_{\text{ref}} \rangle. \quad (3)$$

Symbols.

- u_{μ} : cartesian displacement of the atoms at Γ , $\mu = 3\kappa + \alpha$; M_{μ} the mass of its atom; τ an optional damping time.
- $\langle A, B \rangle = \frac{s}{N} \sum_k \text{Tr}[A(k) B(k)]$, with s the spin degeneracy and N the number of k points.
- $\Sigma[\Delta\rho]$: self energy of the mean field (Hartree, + SEX with HSEX), linear in $\Delta\rho$; $\Sigma = 0$ in IPA.
- g^{file} : the coupling read from the files of EPW (after the removal of the rigid translation if `acoustic_sum_rule`). With `"screened"` it is the screened coupling of DFPT, g^s .
- χ_0 : static independent-particle response, in the Bloch gauge

$$[\chi_0 V]_{nm}(k) = \frac{f_n - f_m}{\varepsilon_n(k) - \varepsilon_m(k)} V_{nm}(k), \quad (4)$$

zero for pairs with the same occupation and for degenerate pairs.

- K^{BO} : the force constants of ph.x (Born–Oppenheimer, they contain the static screening of the electrons).
- $\Pi_{\mu\nu} = \langle g_\mu, \delta\rho_\nu \rangle$: electronic force constants, with $\delta\rho_\nu$ the static response of the electrons of the model to a unit displacement u_ν .
- ρ_{BO} : a second density matrix, propagated with the same $u(t)$, with the mean field and without the laser (only with "dynamic"):

$$i \dot{\rho}_{\text{BO}}(k) = \left[H_0(k) + \Sigma[\rho_{\text{BO}} - \rho_0] + \sum_{\mu} u_{\mu} g_{\mu}(k), \rho_{\text{BO}}(k) \right]. \quad (5)$$

2 What the code actually does

Everything the code computes is listed here. There are two phases: an initialization, done once, which only prepares g and K (and, with the dynamic reference, $\rho_{\text{BO}}(0) = \rho_0$); and the time propagation of Eqs. (1)–(3) with these fixed g and K . No frequency-dependent quantity ($\chi(\omega)$, $\Pi(\omega)$) is ever computed: they appear only in the analysis of section 6.

Initialization (`Lattice::initialize`, once)

1. **Coupling** (`read_coupling`). $g_{\mu}(R_e)$ at $q = 0$ from `prefix.epmatwp`, times `coupling_scale`; Fourier sum to the k points of the simulation, $g_{\mu}(k) = \sum_{R_e} e^{ik \cdot R_e} g_{\mu}(R_e)$; hermitized, $g \leftarrow (g + g^{\dagger})/2$; FFT to the R grid. Check: $\max |H_{\text{EPW}}(k) - H_{\text{tb}}(k)|$ (same Wannier gauge).
2. **Force constants** (`read_dynamical_matrix`). Masses and $C_{\mu\nu}$ of ph.x at Γ ; the acoustic sum rule imposed on C (if `acoustic_sum_rule`); $K \leftarrow (C + C^{\dagger})/2 = K^{\text{BO}}$. Normal modes of K^{BO} , only for the output `Phonon_modes.txt`.
3. **Rigid translation** (`remove_translation_from_coupling`, if `acoustic_sum_rule`). $g_{\kappa\alpha} \leftarrow g_{\kappa\alpha} - \frac{M_{\kappa}}{M_{\text{tot}}} \sum_{\kappa'} g_{\kappa'\alpha}$.
4. **Unscreening** (`unscreen_coupling`, only "screened" with the mean field). For each μ :
 - $\tilde{g}_{\mu} = \chi_0 g_{\mu}$: $U^{\dagger} g_{\mu} U$ (Bloch gauge), each element times $\frac{f_n - f_m}{\varepsilon_n - \varepsilon_m}$ (zero for equal occupations or degenerate pairs), back with $U(\cdot)U^{\dagger}$;
 - $\Sigma[\tilde{g}_{\mu}]$: $\rho_0 + \tilde{g}_{\mu}$ to R , `MeanField::self_energy` (which computes $\Sigma[\rho - \rho_0]$), back to k ;
 - $g_{\mu} \leftarrow g_{\mu} - \Sigma[\tilde{g}_{\mu}]$ (now g_{μ}^{b}); $\tilde{g}_{\mu}$ is kept for the next step.
5. **Static response** (`static_response`, only "static"). For each ν , $\delta\rho_{\nu} = \tilde{g}_{\nu}$ (after the unscreening), or $\chi_0 g_{\nu}$ (IPA), or the Anderson solution of $\delta\rho_{\nu} = \chi_0(g_{\nu} + \Sigma[\delta\rho_{\nu}])$ ("bare" with the mean field). Then $\Pi_{\mu\nu} = \frac{1}{N} \sum_k \text{Tr}[g_{\mu} \delta\rho_{\nu}]$ (`MPI_Allreduce`), $\Pi \leftarrow (\Pi + \Pi^{\dagger})/2$, and $K \leftarrow K - \Pi$. Π and $\tilde{g}$ are then discarded.
6. K is stored in `force_constants_`; from here on g and K never change.

Each stage of the time stepper (`Propagator::derivative`, e.g. 4 times per RK4 step)

The state is $(\rho, u, \dot{u})$, plus ρ_{BO} with the dynamic reference.

1. **Hamiltonian** (`build_hamiltonian`): $H = H_0 + \mathbf{E}(t) \cdot \mathbf{r}$ (or the Peierls phase); $H += \Sigma[\rho - \rho_0]$ in R ; $H(R) += \sum_{\mu} u_{\mu} g_{\mu}(R)$ (`add_coupling`).
2. **Force** (`Lattice::force`), with the same ρ , in R : $F_{\mu} = s \sum_R \sum_{mn} \text{Re}[g_{\mu,mn}(R) (\rho - \rho_{\text{ref}})_{mn}^*(R)]$, equal to $\langle g_{\mu}, \rho - \rho_{\text{ref}} \rangle$ by Parseval; $\rho_{\text{ref}} = \rho_0$, or the current ρ_{BO} .
3. **Electrons**: $\dot{\rho} = -i[H, \rho]$ (+ gradient term without Peierls, + decay if set).

4. **Only dynamic:** the same two steps 1 and 3 for ρ_{BO} , with $\mathbf{E} = 0$, no Peierls phase, no force.

5. **Lattice** (`Lattice::derivative`): $\dot{u} = v$, $\dot{v} = -(Ku + F)/M - (2/\tau)v$.

The force is the full, non-linear $\langle g, \rho - \rho_{\text{ref}} \rangle$ of the propagated ρ : in particular it contains the retarded response of the electrons to $u(t)$ at all orders, which is what the linear-response quantity $\Pi(\omega)$ of section 6 describes.

3 The nine combinations

mean field	coupling	reference	g in H and in F	K	ρ_{ref}	Π (computed once, at the beginning)
off	any	none	g^{file}	K^{BO}	ρ_0	—
off	any	static	g^{file}	$K^{\text{BO}} - \Pi$	ρ_0	$\Pi = \langle g, \chi_0 g \rangle$ (one χ_0 per mode)
off	any	dynamic	g^{file}	K^{BO}	ρ_{BO}	—
on	screened	none	$g^{\text{b}} = g^{\text{s}} - \Sigma[\chi_0 g^{\text{s}}]$	K^{BO}	ρ_0	—
on	screened	static	$g^{\text{b}} = g^{\text{s}} - \Sigma[\chi_0 g^{\text{s}}]$	$K^{\text{BO}} - \Pi$	ρ_0	$\Pi = \langle g^{\text{b}}, \chi_0 g^{\text{s}} \rangle$ (no extra cost: $\chi_0 g^{\text{s}}$ comes from the unscreening)
on	screened	dynamic	$g^{\text{b}} = g^{\text{s}} - \Sigma[\chi_0 g^{\text{s}}]$	K^{BO}	ρ_{BO}	—
on	bare	none	g^{file}	K^{BO}	ρ_0	—
on	bare	static	g^{file}	$K^{\text{BO}} - \Pi$	ρ_0	$\delta\rho = (1 - \chi_0 \Sigma)^{-1} \chi_0 g$ by Anderson mixing, $\Pi = \langle g, \delta\rho \rangle$
on	bare	dynamic	g^{file}	K^{BO}	ρ_{BO}	—

Table 1: Ingredients of Eqs. (1)–(3) for each combination of the parameters. In blue the default. Without the mean field "screened" and "bare" are the same thing ($g^{\text{b}} = g^{\text{s}}$ when $\Sigma = 0$). In orange the only case that needs a self-consistent (iterative) solution.

4 The coupling: what the electrons feel in the static limit

During the propagation the mean field generates, through $\Sigma[\Delta\rho]$, the screening of the displacement by the electrons of the model. For a static displacement the electrons respond with $\delta\rho = \chi_0 V$, where V is the total perturbation, coupling plus induced self energy:

$$V = g + \Sigma[\delta\rho] = g + \Sigma\chi_0 V \implies V = (1 - \Sigma\chi_0)^{-1} g \equiv \varepsilon^{-1} g. \quad (6)$$

The coupling of EPW is already screened by all the electrons of DFT, so with the mean field it must be unscreened first. Asking $V = g^{\text{s}}$ gives the coupling to put in H directly, without iterations:

$$g^{\text{b}} = \varepsilon g^{\text{s}} = g^{\text{s}} - \Sigma[\chi_0 g^{\text{s}}]. \quad (7)$$

5 The adiabatic reference: the frequency of the lattice

K^{BO} already contains the static screening of the lattice by the electrons, $K^{\text{BO}} = K^0 + \Pi$, and the propagated electrons generate it again. The reference decides how this double counting is removed.

mean field	coupling	g in H	static perturbation V felt by the electrons
off	any	g^{file}	g^{file}
on	screened	$g^{\text{b}} = g^{\text{s}} - \Sigma[\chi_0 g^{\text{s}}]$	g^{s} , exactly the one of DFPT
on	bare	g^{file}	$\varepsilon^{-1} g^{\text{file}} = (1 - \Sigma\chi_0)^{-1} g^{\text{file}}$

Table 2: The static perturbation felt by the electrons. From the screened to the bare coupling is a multiplication by ε ; from the bare to the screened one is an inversion of ε .

reference	K	$M\omega^2 \approx$	meaning
none	K^{BO}	$K^{\text{BO}} + \Pi(\omega)$	the screening is counted twice: the lattice is too soft
static	$K^{\text{BO}} - \Pi$	$K^{\text{BO}} + \Pi(\omega) - \Pi(0)$	non-adiabatic phonons (Lazzeri–Mauri 2006); in an insulator K^{BO} up to $(\omega/E_{\text{gap}})^2$
dynamic	K^{BO}	K^{BO} exactly	ρ_{BO} removes the whole response to $u(t)$, also the physical non-adiabatic correction $\Pi(\omega) - \Pi(0)$ of the ground state; twice the cost of the propagation

Table 3: Frequencies of the free lattice (no laser) for the three adiabatic references.

Without laser, at linear order, the response of the electrons to $u(t)$ at the frequency ω gives the force constants $\Pi(\omega)$, so the lattice oscillates with:

With the static reference using $K^{\text{BO}} - \Pi$ and the whole $\rho - \rho_0$ in the force is the same as using K^{BO} and only the excited part $\rho - \rho_0 - \sum_{\nu} u_{\nu} \delta \rho_{\nu}$, since $\langle g_{\mu}, \sum_{\nu} u_{\nu} \delta \rho_{\nu} \rangle = \sum_{\nu} \Pi_{\mu\nu} u_{\nu}$.

6 Static vs dynamic ρ_{BO} , and $\Pi(\omega)$: a linear-response analysis

This section does not describe something the code computes (that is section 2): it linearizes the equations of section 1 without laser, to understand which frequency the lattice gets. The last paragraph discusses which reference is right, also with the laser.

Both references subtract a ρ_{BO} . The force is always $\langle g, \rho - \rho_{\text{BO}} \rangle$ with K^{BO} ; the static reference written as $K^{\text{BO}} - \Pi$ is only a convenient way of doing the same thing. What differs is the ρ_{BO} that is subtracted:

$$\text{static: } \rho_{\text{BO}}(t) = \rho_0 + \sum_{\nu} u_{\nu}(t) \delta \rho_{\nu} \quad \text{depends only on } u \text{ at the same time,} \quad (8)$$

$$\text{dynamic: } \rho_{\text{BO}}(t) \text{ propagated with Eq. (5)} \quad \text{depends on the whole history } u(t' < t). \quad (9)$$

The static one is the Born–Oppenheimer state: electrons infinitely fast, always in the ground state of the displaced lattice (at first order in u). The dynamic one has the finite response time of the electrons ($\sim 1/E_{\text{gap}}$): for $u(t) = u \cos \omega t$ the two are $\delta \rho_{\text{BO}} = \chi(0) g u(t)$ and $\delta \rho_{\text{BO}} = \chi(\omega) g u(t)$. They coincide only for $\omega \rightarrow 0$.

Which one must be subtracted. The subtraction removes what K^{BO} already contains. ph.x is static DFPT, so K^{BO} contains the instantaneous response $\Pi(0)$, which is what the static reference removes; the remaining $\Pi(\omega) - \Pi(0)$ is physics that ph.x does not have (non-adiabatic phonons). The dynamic reference removes $\Pi(\omega)$, i.e. also this physical correction.

Two-level example. One valence and one conduction band, gap Δ , coupling $g_{cv} = g$, IPA. With the convention $u \propto e^{-i\omega t}$, $\delta\rho_{nm} = \frac{f_n - f_m}{\varepsilon_n - \varepsilon_m - \omega - i\eta} (gu)_{nm}$, and

$$\Pi(\omega) = -\frac{2s|g|^2\Delta}{\Delta^2 - \omega^2} = \Pi(0) \left[1 + \frac{\omega^2}{\Delta^2} + \dots \right], \quad \Pi(0) = -\frac{2s|g|^2}{\Delta}. \quad (10)$$

The dynamic ρ_{BO} is larger than the static one by $1 + (\omega/\Delta)^2$: in an insulator with $\omega \ll E_{\text{gap}}$ the two references are equivalent up to this factor (for the real hBN, 1351.2 vs 1349.5 cm⁻¹).

Many bands. Each valence–conduction pair at each k is a two-level system with its own transition energy $\Delta_{cv}(k) = \varepsilon_c(k) - \varepsilon_v(k)$. Collecting the pairs (c, v) and (v, c) (g hermitian at Γ),

$$\Pi_{\mu\nu}(\omega) = -\frac{s}{N} \sum_k \sum_{v,c} \frac{2\Delta_{cv}(k) \text{Re}[g_{\mu,vc}(k) g_{\nu,cv}(k)]}{\Delta_{cv}(k)^2 - \omega^2} \quad (11)$$

(plus an imaginary part antisymmetric in $\mu\nu$, which vanishes after the sum over k with time-reversal symmetry). For small ω ,

$$\Pi_{\mu\nu}(\omega) - \Pi_{\mu\nu}(0) \simeq -\omega^2 \frac{s}{N} \sum_k \sum_{v,c} \frac{2 \text{Re}[g_{\mu,vc} g_{\nu,cv}]}{\Delta_{cv}(k)^3} : \quad (12)$$

an average of $(\omega/\Delta_{cv})^2$ over the transitions, weighted by $|g|^2/\Delta$, dominated by small gaps with large coupling. For ω above the smallest $\Delta_{cv}(k)$ the poles give $\Pi(\omega)$ an imaginary part: the phonon decays into electron–hole pairs. With the mean field the poles are the excitations Ω_λ of the model (excitons with HSEX) instead of the $\Delta_{cv}(k)$: $\Pi(\omega) = -\sum_\lambda 2\Omega_\lambda |\langle g, \lambda \rangle|^2 / (\Omega_\lambda^2 - \omega^2)$. Bands outside the Wannier model enter only statically, through g^s and K^{BO} .

$\Pi(0)$ in the code. A $3N_{\text{at}} \times 3N_{\text{at}}$ matrix (Ha/bohr²) returned by `Lattice::static_response`, computed once in `Lattice::initialize` and immediately subtracted from K :

```
auto Pi = static_response(...); K(mu, nu) -= Pi(mu, nu);
```

It is not kept: afterwards it exists only inside `force_constants_ = KBO -  $\Pi(0)$` . The recap prints the frequencies of $K^{\text{BO}} - \Pi(0)$ next to those of `ph.x`, and the residual of the acoustic sum rule of Π ; Π itself is not written to any file.

$\Pi(\omega)$ in the code. It does not exist: no variable or routine computes it. It is the name, in linear response, of what the propagation of section 2 does: $F(t)$ contains the response of the electrons to the whole history of u ; linearized,

$$F_\mu(t) = \sum_\nu \int dt' \Pi_{\mu\nu}(t - t') u_\nu(t') \xrightarrow{u \propto e^{-i\omega t}} F = \Pi(\omega) u, \quad (13)$$

with, in IPA, $\Pi_{\mu\nu}(\omega) = \frac{s}{N} \sum_k \sum_{nm} g_{\mu,mn} g_{\nu,nm} \frac{f_n - f_m}{\varepsilon_n - \varepsilon_m - \omega - i\eta}$, and with the mean field $\chi_0(\omega) \rightarrow \chi(\omega) = (1 - \chi_0(\omega)\Sigma)^{-1}\chi_0(\omega)$ (TDHF response of the model, which the linearized Eq. (1) solves without laser). The code always uses the full F , also beyond linear order and with the electrons excited by the laser.

How to see it. $\Pi(\omega)$ can only be extracted afterwards: from a run without laser and a small `initial_displacement`, the frequency ω of $u(t)$ in `Lattice.txt` satisfies $M\omega^2 = K^{\text{BO}} - \Pi(0) + \Pi(\omega)$, so, projected on the mode, $\Pi(\omega) - \Pi(0) = M\omega^2 - K^{\text{BO}}$. This is what the “static limit” check of `ci-test/check_phonons.py` measures, comparing ω with `ph.x`.

contribution of the electrons to the force	in K^{BO} ?	static	dynamic
$\Pi(0)$: static response of the ground state, linear in u	yes	removed ✓	removed ✓
$\Pi(\omega) - \Pi(0)$: non-adiabatic response of the ground state	no	kept ✓	removed ✗
response of the ground state non-linear in u	no (K^{BO} is harmonic)	kept ✓	removed ✗
response of the electrons excited by the laser	no	kept ✓	kept ✓

Table 4: What each adiabatic reference removes from the force of the electrons. A ✓ means the right choice: remove what is in K^{BO} , keep the rest.

Which reference is better, also with the laser. The criterion is the one above: remove only what K^{BO} already contains. K^{BO} is static (DFPT) and harmonic (second derivative of the energy at equilibrium), so it contains exactly one electronic contribution, the static response of the ground state at linear order in u , $\Pi(0)$. Everything else is physics that ph.x does not have and must be kept. With the laser, ρ contains also the response of the excited electrons to u ; ρ_{BO} never sees the laser, so this part is kept by both references.

The dynamic ρ_{BO} is the whole response of the ground state to $u(t)$, at all frequencies and all orders, so it removes more than what K^{BO} contains: the non-adiabatic correction and the anharmonic response of the electrons of the model, which are not double counted with the static reference, because ph.x has neither of them. **The static reference is therefore the right one also with the laser.** In practice the difference is small in an insulator: $(\omega/E_{\text{gap}})^2$ for the non-adiabatic part, and the non-linear terms matter only for large displacements; on the part that matters with the laser, the response of the excited electrons, the two references coincide. The dynamic reference is useful as a test (without laser it gives exactly K^{BO}) and as a cross-check of the static one (a large difference means that the non-adiabatic correction is not small); it doubles the cost of the propagation.

7 How Π is computed

Only with the static reference. The static response $\delta\rho_\nu$ solves

$$\delta\rho_\nu = \chi_0(g_\nu + \Sigma[\delta\rho_\nu]), \quad (14)$$

and then $\Pi_{\mu\nu} = \langle g_\mu, \delta\rho_\nu \rangle$, symmetrized, $\Pi \leftarrow (\Pi + \Pi^\dagger)/2$.

mean field	coupling	$\delta\rho_\nu$	$\Pi_{\mu\nu}$	cost
off	any	$\chi_0 g_\nu$	$\langle g_\mu, \chi_0 g_\nu \rangle$	one χ_0 per mode
on	screened	$\chi_0 g_\nu^s$	$\langle g_\mu^b, \chi_0 g_\nu^s \rangle$	none: $\chi_0 g^s$ is already computed by the unscreening (one χ_0 and one Σ per mode)
on	bare	$(1 - \chi_0 \Sigma)^{-1} \chi_0 g_\nu$	$\langle g_\mu, \delta\rho_\nu \rangle$	Anderson mixing, one χ_0 and one Σ per iteration and per mode

Table 5: The static response and Π . With "screened" the solution is known without iterations because $g^b + \Sigma[\chi_0 g^s] = g^s$: the total perturbation is g^s by construction.

Why Anderson mixing exists. Only in the combination mean field + "bare" + "static" (orange in Table 1). There the starting point is the bare coupling, and the screened response requires ε^{-1} : a linear system of size $N n_b^2$, solved iteratively because the plain iteration $\delta\rho \leftarrow \chi_0(g + \Sigma[\delta\rho])$ does not

converge with a strong mean field. The option "bare" was introduced first, for the synthetic tests, where g is bare by construction; with a real EPW run ("screened") the self-consistency is never needed.

Sign. In an insulator $f_n - f_m$ and $\varepsilon_n - \varepsilon_m$ have opposite signs for each valence–conduction pair, so χ_0 and Π are negative: $K^0 = K^{\text{BO}} - \Pi$ is harder than K^{BO} , as expected (the electrons soften the lattice).